

Data Multimodal Film Genre Classification CNN and LSTM Models on the IMDB Dataset

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1.Introduction

This report compares two deep learning models trained to guess movie genres using a slice of IMDB data. One is a CNN that looks at movie posters, and the other is an LSTM that reads plot summaries. Since a film can be a comedy, horror, and romance all at once, this is a multi-label problem—which is a lot messier than just picking one category.

In the sections ahead, we'll dig into how each model actually trained, whether they were just memorizing or actually learning patterns, and how accurate their predictions ended up being on films they'd never seen before.

2. CNN — Poster-Based Classification

2.1 Training Dynamics

The CNN ran through the whole dataset fifty times (that's the 50 epochs). Figure 1 shows the curves for loss, precision, and recall as training progressed, giving us a clear picture of how the model improved round by round.

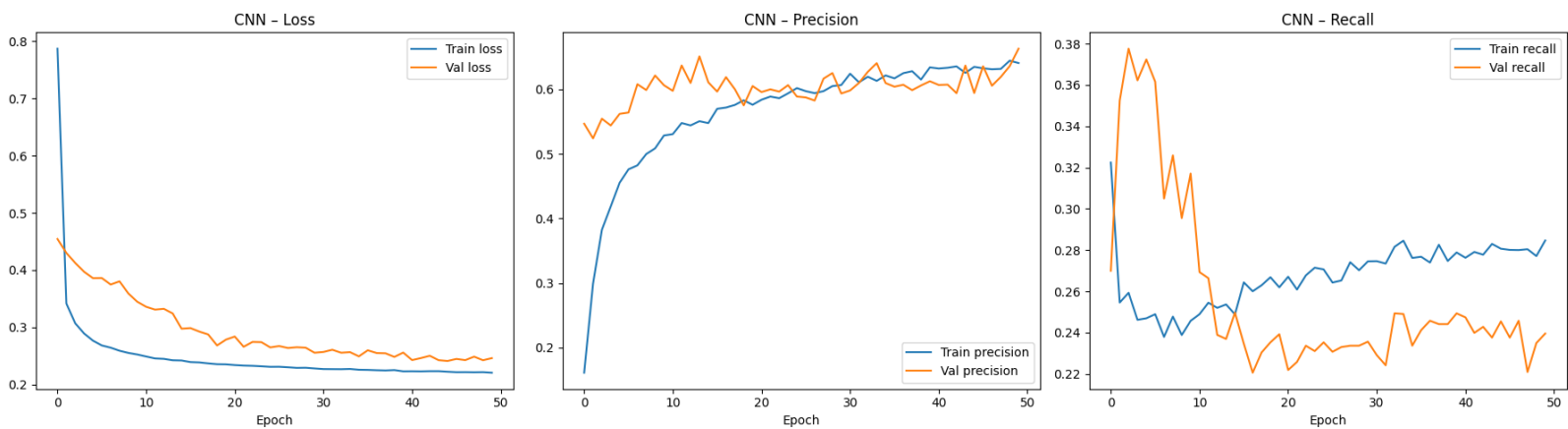


Figure 1: CNN training and validation Loss, Precision, and Recall over 50 epochs.

Loss: The training loss dropped fast—plummeting from around 0.78 to under 0.30 in the first ten epochs. After that, both the training and validation losses settled into a tight range between about 0.21 and 0.24. That close match is a healthy sign; it means the model isn't just memorizing the training posters but is learning patterns that hold up on new images. This robustness comes from how convolutional layers work, focusing on small, local patches of an image rather than the whole thing at once.

Precision: When the model did decide to predict a genre, it was right roughly two-thirds of the time. Validation precision started off higher than training precision early on—a common quirk when the model is cautious and makes few positive guesses. Both curves eventually climbed to around 0.65–0.68 by the fiftieth epoch, though the validation line showed some jagged ups and downs. That's expected with a smaller validation set and the complexity of multiple possible labels per movie.

Recall: Here's where things get shaky. The model only managed to catch about a quarter of all the actual genres present in the films. Training recall flatlined around 0.28, while validation recall—after an early jump to 0.37—settled back down to roughly 0.22–0.25. That wide gap between decent precision and poor recall tells us the model plays it way too safe. It only speaks up when it's extremely confident, leaving a huge chunk of true labels unnoticed. And honestly, it's hard to blame it. Drama and Romance posters often look nearly identical: moody close-ups, soft lighting, warm tones. Without text or context, telling them apart visually is a tall order.

2.2 Qualitative Analysis of Predictions

The CNN slaps Drama on almost everything—just because it saw so much of it during training. For Hannibal Rising, it guessed Horror instead of Crime, which kinda makes sense with that dark, moody poster. Twilight got tagged as Horror and Thriller because that pale figure in the shadows just looks scary, even though the story isn't. Comedy is where it really bombs: The Out-of-Towners came back as Drama, Thriller, Action—way off. Though occasionally, when a comedy poster is super obvious and old-school, it actually gets it right. Basically, the model judges the book by its cover, and sometimes the cover lies.

3. LSTM — Overview-Based Classification

3.1 Training Dynamics

The LSTM was trained for 20 epochs. Figure 2 shows loss and accuracy throughout training.

The model clearly shows signs of overfitting. While the training loss steadily improves from about 0.35 down to 0.10, the validation loss starts increasing after around epoch 4 and eventually reaches roughly 0.30 by epoch 19. At the same time, training accuracy rises to

about 56%, but validation accuracy stays unstable and levels off between 38% and 42%, with the best performance occurring around epoch 5. This pattern suggests that the model is learning the training data too well but struggling to generalise to unseen data. Stopping the training early, around epoch 4 or 5, would likely have produced a better-performing model.

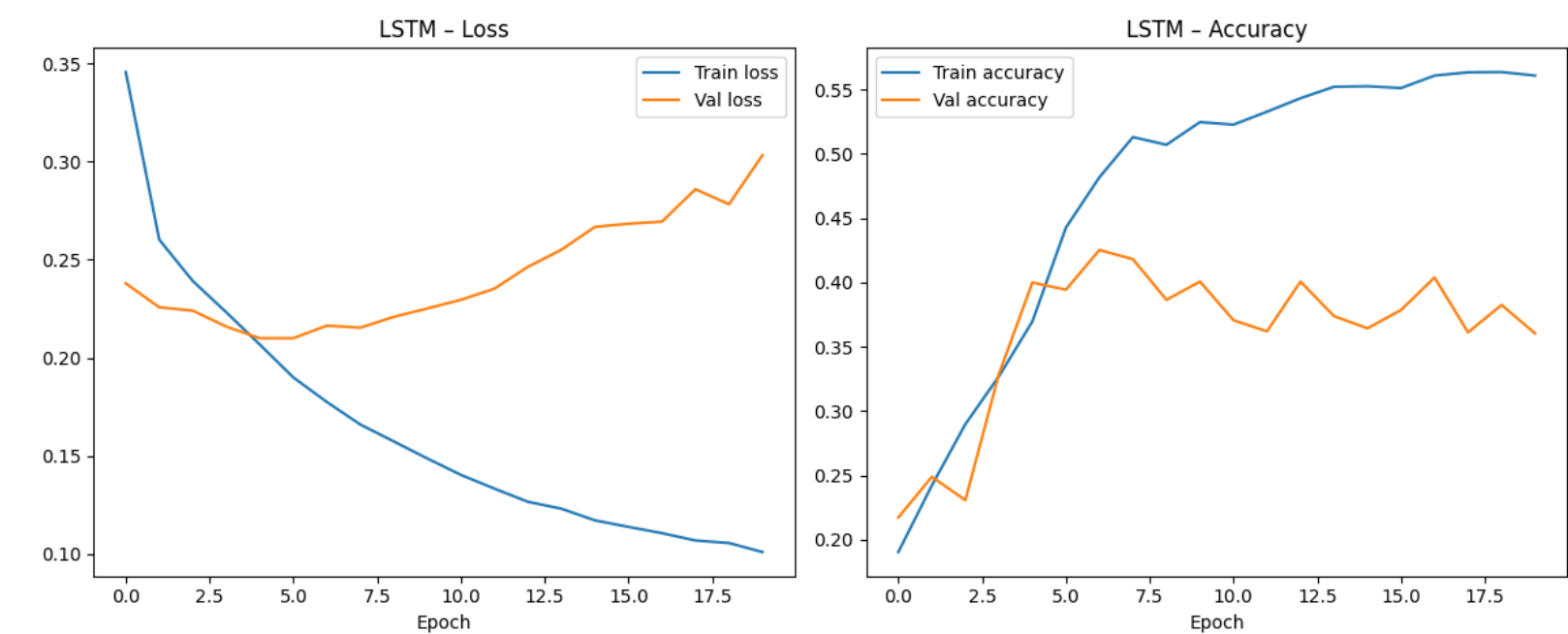
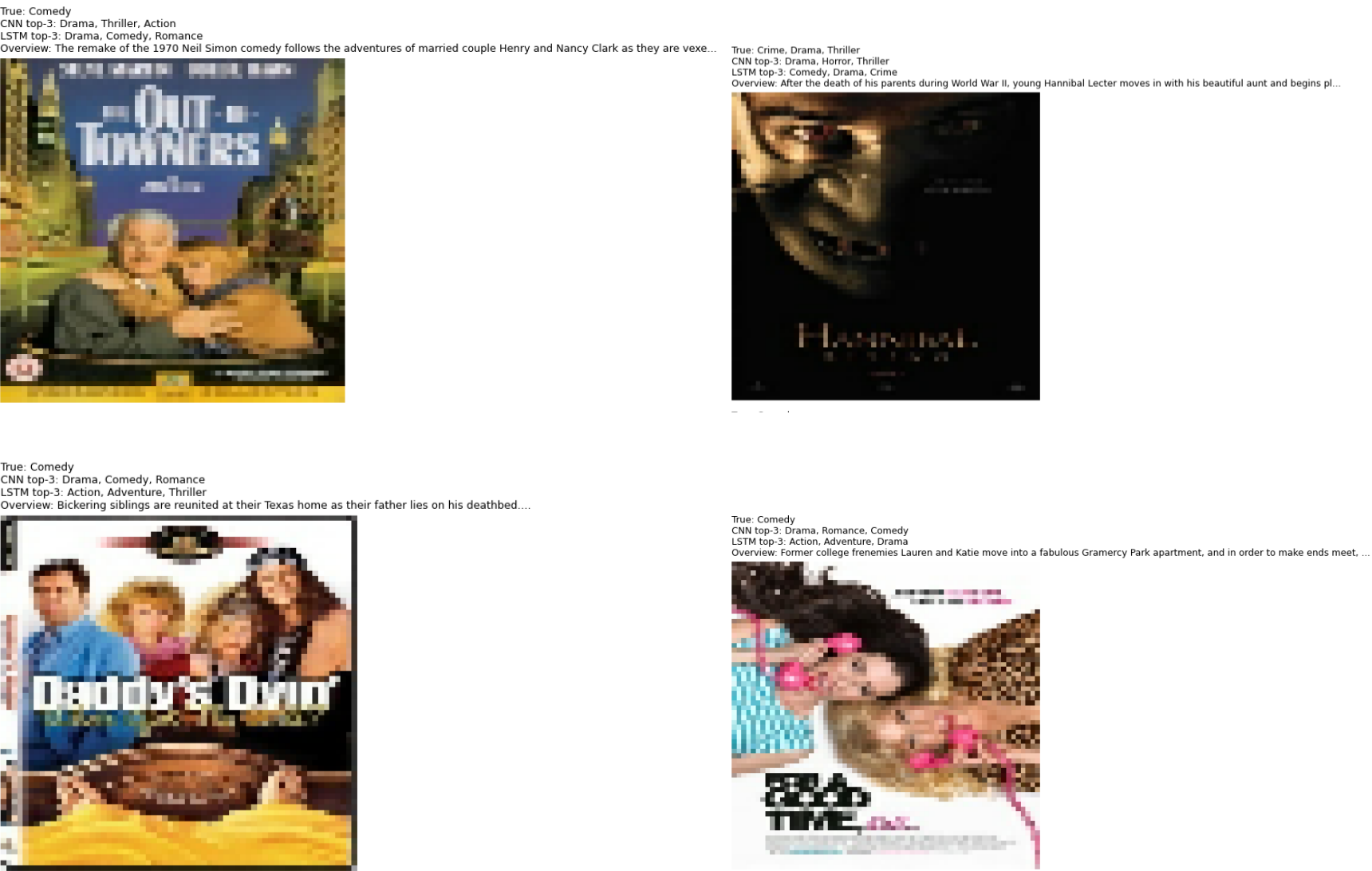


Figure 2: LSTM training and validation Loss and Accuracy over 20 epochs.

One key reason for this is that the embedding layer is being trained from scratch on a relatively small dataset. This means the model has to learn both word meanings and genre patterns at the same time, which is difficult with limited data. Without using pre-trained embeddings like GloVe or FastText, or techniques such as recurrent dropout, the model tends to rely on patterns that are specific to the training set and don’t generalise well. Additionally, movie overviews within the same genre can vary a lot in terms of vocabulary and writing style, making it even harder for the model to learn consistent patterns across the data.

3.2 Qualitative Analysis of Predictions

Figure 3 shows example model predictions alongside posters and overview text.



The LSTM does a good job when it can rely on clear, genre-specific words in the text. For example, in Hannibal Rising (Crime, Drama, Thriller), it correctly picks up on “Crime” because the description includes obvious references to criminal activity. However, it also wrongly predicts “Comedy,” probably because some of the early parts of the overview use family-related language that can be misleading. Similarly, for Twilight (Drama, Fantasy, Romance), the model correctly identifies “Drama” and “Romance” thanks to the strong focus on relationships and emotions. It misses “Fantasy,” though, likely because words like “vampire” or “immortal” aren’t well understood in its learned vocabulary.

True: Action, Horror, Thriller
CNN top-3: Drama, Thriller, Action
LSTM top-3: Drama, Comedy, Action
Overview: When the crew of an American tugboat boards an abandoned Russian research vessel, the alien life form aboard regards the...



True: Drama, Fantasy, Romance
CNN top-3: Drama, Horror, Thriller
LSTM top-3: Drama, Comedy, Romance
Overview: Bella Swan moves to Forks and encounters Edward Cullen, a gorgeous boy with a secret....

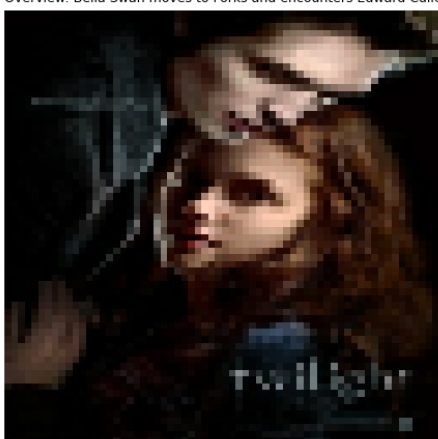


Figure 3: Example predictions for CNN (top-3) and LSTM (top-3) vs. true genre labels

Where the model struggles is with genres that don't have clear or typical keywords. For instance, Daddy's Dying (Comedy) is completely misclassified as Action and Thriller. This happens because the story revolves around a serious family situation, and the language used sounds more dramatic than humorous. In another case, an alien-crew film (Action, Horror, Thriller) is predicted as Drama, Comedy, and Action. While it does manage to get "Action" right, it misses the horror and sci-fi elements because the language overlaps a lot with general dramatic descriptions, making it harder for the model to distinguish the true genre.

4. Comparative Assessment and Recommendations

Generalisation:

The CNN generalises much better than the LSTM. Its training and validation loss stay close to each other over all 50 epochs, showing that it is learning patterns that work well on new data. In contrast, the LSTM starts to overfit quite early—around epoch 5—as its validation performance begins to drop. This is also reflected in the results: the CNN achieves a validation precision of about 0.67, while the LSTM only reaches around 0.40 accuracy.

Class Imbalance:

Both models tend to predict "Drama" when the input is unclear, which suggests that the dataset is imbalanced. Since Drama appears more frequently, the models become biased toward it. Using techniques like class-weighted loss functions or oversampling minority genres could help reduce this bias and improve performance, especially for less common genres.

Recommended Improvements:

For the CNN, using transfer learning—such as models pre-trained on ImageNet like EfficientNet or ResNet50—would likely improve performance by providing richer visual features instead of learning everything from scratch.

For the LSTM, replacing the current embeddings with pre-trained ones like GloVe or FastText, along with adding recurrent dropout, would help reduce overfitting and improve generalisation.

Finally, combining both models in a multimodal approach could give the best results. By merging predictions from the CNN (images) and LSTM (text), the system can use both visual and textual information, which would likely lead to better overall performance than using either model on its own.

5. Conclusion

Both models show that it is possible to classify film genres from poster images and plot summaries, but it's not an easy task. The CNN performs more consistently and generalises well to new data, although it tends to be cautious in its predictions, which leads to lower recall. On the other hand, the LSTM is good at picking up genre-related words from the text, but it struggles with overfitting due to limited data and a lack of proper regularisation.

Both models perform best on the dominant "Drama" class and find it harder to correctly identify less common genres, especially when the visual or textual cues are unclear or overlapping. Overall, the most effective ways to improve performance would be to handle class imbalance, use pre-trained representations, and apply better regularisation techniques.